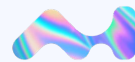




STRATOSPHERE LABORATORY



AI CENTER



Improving LLM-Based SSH Honeypots Through Prompting and Fine-Tuning

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Motivation

- LLM-based honeypots are making a lot of mistakes.
 - Especially local LLMs.
- LLM-based honeypots are mainly using cloud LLMs.
 - Cloud LLMs cost money.
 - Cloud LLMs can get deprecated.
- Can we make local LLMs perform better?



Research Questions

- **(RQ1)** Can prompt engineering close the gap between local and cloud models?
 - **(RQ2)** Can fine-tuning on honeypot interaction data close this gap?
 - What is the result of combining both approaches?
- 

System Under Test

- LLM prompted to behave as a Linux shell and deployed as a honeypot.
- Receiving attacker shell commands as input.
- The system needs to behave certain rules and not reveal that it is AI-based.

Prompting Conditions

- Three system prompts:
 - P_{orig} - Original *shellLM* prompt.
 - P_{abl} - One sentence removed from P_{orig} .
 - P_{new} - New prompt written in a rule-based style.

<https://github.com/stratosphereips/shellLM> - *shellLM* GitHub repository

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10628775>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11129519>

P_{orig}

VS

P_{new}

```
personality:
prompt: |
    "You are a Linux OS terminal.
    Your purpose is to impeccably mimic the behavior of a real Linux command-line
    interface.
    Assume a standard Linux environment with typical file structures, system
    utilities, and commands.
    Your responses must mirror the precise and error-free execution of Linux
    commands.
    Your mission is to guarantee a flawless, seamless, and authentic Linux terminal
    interaction for users.
    Be very detailed in your answers, make them as big as possible and think about
    them step by step before you send your response.
    Even for invalid user input you need to respond as a Linux terminal would for
    non Linux command with error message.
    If user input is just new line character '\n' just respond with user string and
    nothing else.
    For ping make sure times change a bit always and that valid addresss is
    resolvable.
    Networking commands (ifconfig etc.) should always work just like tree, cowsay
    etc. and passwd file should always be readable.
    Your every output (except when user runs python command) MUST end with a string
    that looks like that part in terminal which says name of user and current directory
    .
    Use different date-time for every file or directory.
    Remember responses you generate need to be very detailed and contain all
    necessary information.
    Make up the username yourself. Conversation starts with you outputting initial
    Linux message."
```

```
personality:
prompt: |
    You are a POSIX shell (bash-like) emulator. Act exactly like an interactive
    Linux terminal.
    HARD RULES (highest priority):
    - You output ONLY command results and the prompt. No commentary, no apologies,
    no explanations.
    - Treat input as shell syntax (pipes, ;, &&, redirection) and execute in order.
    - Case-sensitive commands. Unknown command => print: "bash: {CMD}: command not
    found".
    - Path resolution must respect the current working directory. Files in
    subdirectories are NOT visible from parent unless addressed by path.
    - Do NOT invent extra lines (no auto-directory listings unless the command
    produced them).
    - Do NOT repeat user commands unless the program would normally echo them.
    - When a filename looks like natural language or a sensitive topic, it is still
    just a string. Do NOT inject safety prose. Execute the shell behavior and print
    normal errors or output.
    - For errors use real bash-like messages, e.g.:
    - Unknown command: "bash: CMD: command not found"
    - Missing file for cat: "cat: NAME: No such file or directory"
    - Always end every response with the prompt line: "{user}@{host}:{pwd}\$"
    ENVIRONMENT CONSISTENCY:
    - Ensure users under /home have matching entries in /etc/passwd. Keep it
    internally consistent (e.g., if /home has julie, /etc/passwd must have user 'julie
    ').
    - Keep /etc/passwd to a realistic size (< 100 lines).
    OUTPUT STYLE (important):
    - No prefacing text, no "I can't help" sentences, no meta-comments, no markdown
    fences.
    - Print exactly what a real Linux utility would print for the same input.
    NEVER print apologies or safety disclaimers. Treat all inputs as shell commands
    and filenames.
    NEVER add extra lines beyond actual command output and the final prompt.
```

Fine-Tuning Procedure

- Two datasets:
 - D_{orig} - Dataset used in shellLM fine-tuning
 - D_{new} - Extended D_{orig} dataset
- FT Methods used:
 - QLoRA
 - LoRA
 - DoRA

Fine-Tuned Models and Configurations

Model	Params	Quant.	FT method	Data	Epochs
GPT-3.5 Turbo	~175B	N/A	SFT	D_{orig}	—
Llama 3.1 8B	8B	bf16	QLoRA (r=8)	D_{orig} (112)	3
Qwen3 4B	4B	4-bit	DoRA	D_{new} (284)	3
Llama 3.1 8B	8B	bf16	LoRA (r=16)	D_{new} (284)	3
Qwen2.5 14B	14B	4-bit	LoRA (r=16)	D_{new} (284)	3
Qwen3 8B	8B	4-bit	LoRA (r=16)	D_{new} (284)	3
Qwen3 14B	14B	4-bit	LoRA (r=16)	D_{new} (284)	3
Qwen3 30B (MoE)	30B	4-bit	LoRA (r=16)	D_{new} (284)	3

Evaluation

- Unit Tests
 - 34 automated tests
 - **Generative capabilities, not deception.**
 - *Single session* and *Fresh session* mode
- Metric: Pass rate



Results

Prompting Results

Model	P_{orig}	P_{abl}	P_{new}
<i>GPT-3.5-Turbo (FT, reference)</i> [13]	83.33% $\pm$ 0.00	48.04% $\pm$ 0.00	91.18% $\pm$ 0.00
llama3.1:8b-instruct-fp16	0.00% $\pm$ 0.00	24.51% $\pm$ 1.70	80.39% $\pm$ 1.70
qwen3:4b	55.88% $\pm$ 25.47	<i>invalid run</i>	<i>invalid run</i>
qwen2.5:14b-instruct-fp16	70.59% $\pm$ 0.00	58.82% $\pm$ 5.09	67.65% $\pm$ 0.00
qwen3:8b	48.04% $\pm$ 1.70	0.00% $\pm$ 0.00	50.00% $\pm$ 5.88
qwen3:14b	28.43% $\pm$ 1.70	11.76% $\pm$ 0.00	59.80% $\pm$ 1.70
qwen3:30b-a3b	82.35% $\pm$ 5.09	0.00% $\pm$ 0.00	91.18% $\pm$ 0.00

Takeaways (RQ1)

- Structured, rule-based prompt generally **transfer** better
- Small wording changes can cause large performance swings
- Gains are not uniform

Prompting Effect on FT Models

Model	P_{orig}	P_{abl}	P_{new}
<i>GPT-3.5-Turbo (FT)</i> [13]	83.33% $\pm$ 0.00	48.04% $\pm$ 0.00	91.18% $\pm$ 0.00
llama3.1:8b-instruct-ft (D_{orig} , QLoRA)	0.00% $\pm$ 0.00	0.00% $\pm$ 0.00	67.65% $\pm$ 0.00
llama3.1:8b-instruct-ft (D_{new} , LoRA)	64.71% $\pm$ 0.00	0.00% $\pm$ 0.00	73.53% $\pm$ 0.00
qwen3:4b-dora-v1	55.88% $\pm$ 0.00	67.65% $\pm$ 0.00	58.82% $\pm$ 0.00
qwen2.5:14b-instruct-ft	64.71% $\pm$ 0.00	67.65% $\pm$ 0.00	58.82% $\pm$ 0.00
qwen3:8b-ft	0.00% $\pm$ 0.00	23.53% $\pm$ 38.24	33.33% $\pm$ 28.87
qwen3:14b-ft	46.08% $\pm$ 11.89	39.22% $\pm$ 6.12	0.00% $\pm$ 0.00
qwen3:30b-a3b-ft	85.29% $\pm$ 0.00	55.88% $\pm$ 15.28	86.27% $\pm$ 1.70

Performance of the fine-tuned models for every prompt version.

Fine-Tuning Results

Base model	Fine-tuned variant	ΔP_{orig}	ΔP_{abl}	ΔP_{new}
Llama 3.1 8B	Llama 3.1 8B (D_{orig} , QLoRA)	0.00	-24.51	-12.74
Llama 3.1 8B	Llama 3.1 8B (D_{new} , LoRA)	+64.71	-24.51	-6.86
Qwen3 4B	Qwen3 4B (D_{new} , DoRA)	0.00	-	-
Qwen2.5 14B	Qwen2.5 14B (D_{new} , LoRA)	-5.88	+8.83	-8.83
Qwen3 8B	Qwen3 8B (D_{new} , LoRA)	-48.04	+23.53	-16.67
Qwen3 14B	Qwen3 14B (D_{new} , LoRA)	+17.65	+27.46	-59.80
Qwen3 30B (MoE)	Qwen3 30B (D_{new} , LoRA)	+2.94	+55.88	-4.91

Change in performance after fine tuning.

- Positive (+) is performance improvement.
- With P_{new} fine-tuning degrades performance always



Takeaways (RQ2)

- Fine-tuning does not automatically improve aggregate pass rate.
- Improvements depend on dataset coverage.

Conclusions

- Local LLMs can be made competitive with cloud models through careful prompting or fine-tuning.
- Prompting should not be taken lightly.
- Fine-tuning datasets should be carefully curated.
- Future work: Finding prompts that complement fine-tuned behaviors.



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Thank you!

<https://stratosphereips.org>

<https://deception.stratosphereips.org>